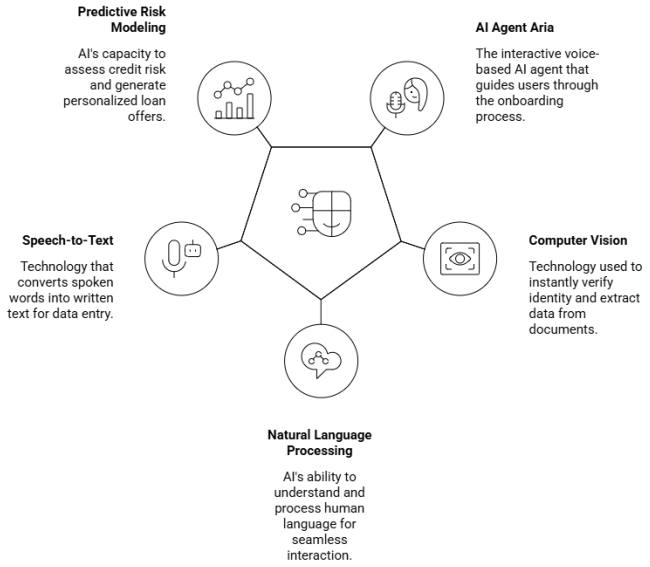
TensorX

STRATEGIC SUBMISSION 2026

TensorX

Pioneering Agentic AI Architecture for the Next Generation of Poonawalla Fincorp Digital Lending.

TensorX Platform Architecture



Made with Napkin

Executive Brief

TensorX is not a form-filler; it is a full-stack loan underwriting swarm.

10.6s

DECISION VELOCITY

Complete onboarding from first "Hello" to final "Sanction Letter".

96%

OPEX REDUCTION

Eliminating manual KYC processing and physical verification loops.

75+

AI AGENTS

A swarm of specialized micro-models vetting every applicant signal.

Market Challenges

Identifying the systemic friction in modern lending.

67% User Attrition

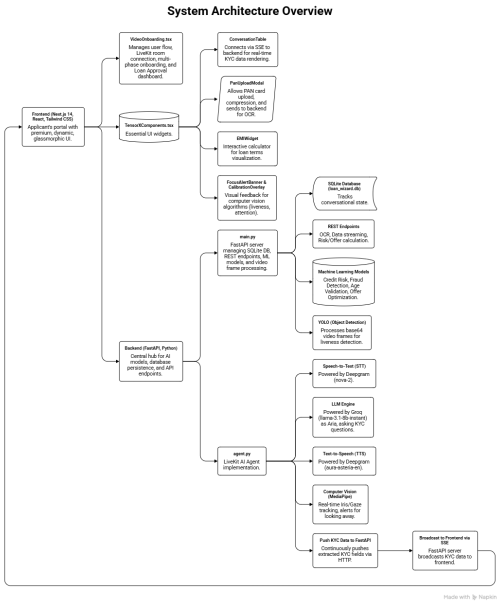
Static forms and multi-day cycles cause massive dropout in prime and near-prime segments.

Synthetic Fraud

Traditional KYC is blind to sophisticated AI-generated IDs and deepfake video injections.

Regulatory Overhead

Manual RBI V-CIP compliance is expensive and prone to human error during auditing.



Aria: Agentic Voice-First KYC

Moving from form-based entry to trust-based conversation.

Multimodal Verification

Real-time Transcription

Whisper-powered extraction of PAN, Aadhaar, and Income intents.

Continuous Liveness

YOLOv11s monitoring for gaze, lip-sync, and iris movement.

Frictionless Sanction

Instant credit scoring without a single manual keyboard input.

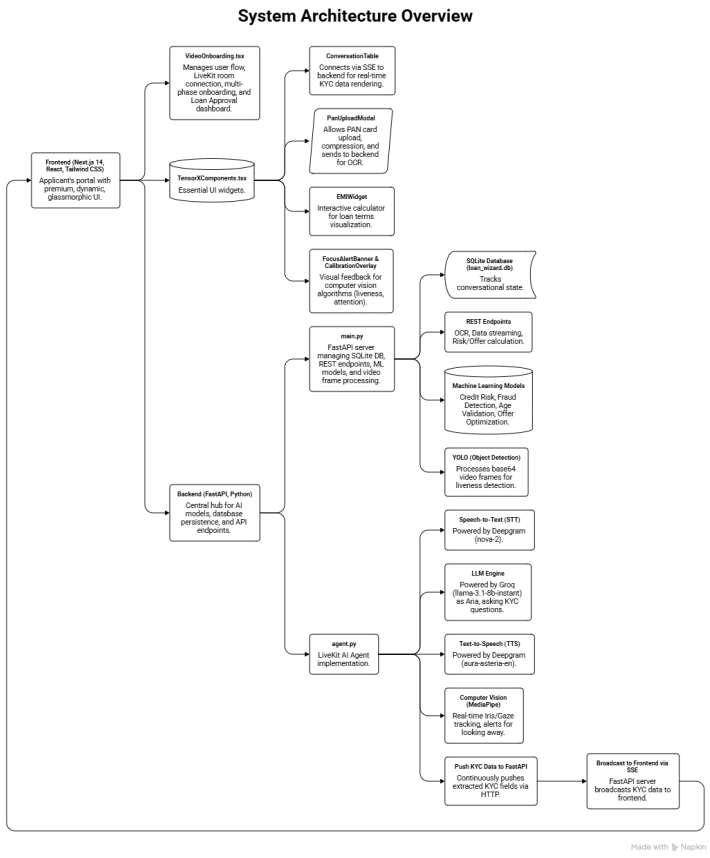
The Competitive Edge

How TensorX outpaces traditional lending stacks.

Feature	Traditional Lending	Neo-Bank Apps	TensorX Agentic AI
Onboarding Time	3–5 Days	10–15 Minutes	**10.6 Seconds**
KYC Method	Manual/Static	Static OCR	**Conversational/Real-time**
Fraud Defense	Reactive	Basic Liveness	**Continuous Biometric Gaze**
Decisioning	Human Underwriter	Rule-based engine	**XGBoost Swarm + SHAP**

Engineering Mastery

A sub-100ms pipeline for enterprise-grade lending.



LiveKit SFU
Stateless WebRTC infrastructure for high-concurrency video sessions.

FastAPI Orchestrator
Async pipeline managing 75 parallel model inferences.

The Swarm Architecture

75+ specialized agents vetting every applicant signal.

Vision Swarm

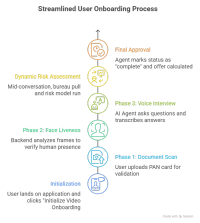
Iris tracking, emotion radar, and pixel-level deepfake CNN protection.

Voice Swarm

Multilingual intent normalization and semantic stress analysis.

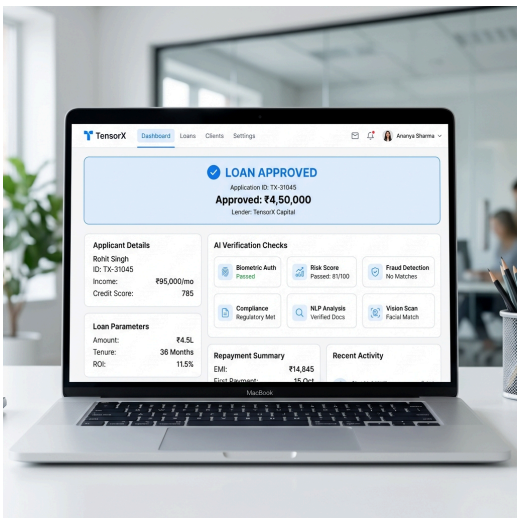
Intelligence Swarm

XGBoost Risk Scoring with full SHAP local explainability for audits.



Business Performance

Quantifying the Poonawalla Fincorp advantage.



99.2%

KYC SUCCESS RATE

10.6s -98% TIME

SESSION DURATION

₹45 SAVING ₹1,155

ONBOARDING COST

The Road Ahead

Scaling the future of frictionless fintech.

Trust & Compliance

RBI V-CIP 2024: Full alignment with latest biometric video norms.

Future Integration

Account Aggregator: Direct API integration for thin-file analysis.

Thank You.

TensorX: Redefining Digital Lending for Poonawalla Fincorp.

View Repository

BUILT TO WIN. BUILT FOR PFL.